

Ori Mosenzon

AI Engineer · Agentic Systems · Deep ML Researcher

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SUMMARY

ML researcher and engineer with 25+ years of experience across research, industry, and teaching. My background spans deep learning, NLP, information theory, and algorithm development. In recent years I've shifted toward building AI-driven applications using agentic workflows — designing systems where LLM agents handle implementation while I focus on architecture and problem framing.

I have strong mathematical foundations and hands-on experience with modern LLM tooling (fine-tuning, RAG, agent orchestration). I currently build AI-driven applications independently and for clients.

CORE COMPETENCIES

Domain	Skills
Agentic AI	LLM orchestration, multi-agent pipelines, Claude Code, prompt engineering, system design
Deep ML	NLP, fine-tuning (GPT-2/LLaMA), transformer architecture, prosody modeling, information theory
Foundations	Information Bottleneck theory, mutual information, algebra, calculus, probabilistic models
Engineering	Python, PyTorch, TensorFlow, Flask, REST APIs, Linux
Applied domains	Conversational AI, audio/speech processing, image processing, financial algorithms

PROFESSIONAL EXPERIENCE

Independent AI Engineer & Consultant

2021 – Present

Designing and building AI-powered applications using agentic workflows — real client work and research projects:

- Inbal** — full-stack management platform for a paying client (scheduling, payments, student management) · Next.js · TypeScript · Prisma · NextAuth
- Sidekick** — AI-powered integration debugging agent with visual browser automation · Flask · Selenium · Claude API · Pillow
- Golem** — open-source VS Code extension with RTL support and Claude-powered agentic logic · TypeScript · VS Code API · Anthropic SDK
- WikiVoice** — automatic Wikipedia article narration with natural prosody modeling · Python · Kokoro-82M TTS
- Karaoke Sync** — real-time lyrics synchronization using speech recognition · Flask · yt-dlp · Whisper · HTML5

AI Engineer & Researcher — KamiComputing

2019 – 2021

Built a conversational AI companion product — an interactive chatbot with personality and natural-sounding speech, aimed at elderly users.

- Fine-tuned GPT-2 for dialogue generation and personality modeling
 - Researched and implemented prosody generation for speech synthesis
 - Worked on semantic understanding and long-context coherence
 - Built before the current wave of LLM assistants, when these problems required custom solutions
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Freelance Deep ML Engineer

2014 – 2019

Applied deep learning to real-world industrial problems across multiple domains:

- **Zeekit** — deep learning for garment fitting and virtual try-on (computer vision)
 - **DEEP** — ML research and algorithm development
 - **Applied Materials** — image processing and algorithmic work in semiconductor manufacturing
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Independent Developer & Math Educator

2008 – 2016

- Built interactive tools for learning linear algebra and high school mathematics
 - Taught programming (C, C++, Python) at Dan Academic School
 - Private mathematics tutoring
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Mathematical Finance Developer — Superderivatives Ltd., Tel Aviv

2007 – 2008

Algorithmic development and software architecture for financial derivatives pricing. Translated formal mathematical models by Quants (financial mathematicians) into production code.

Software Developer — Intel Corp., Haifa

2005 – 2007

Developed design automation tools in C++. Implemented circuit mapping algorithms and extended an HDL compiler.

Lecturer & Programmer — Hebrew University & Hadassah College, Jerusalem

1997 – 2005

Taught the C++ Programming Workshop in the Computer Science department for 8 years.

Algorithm Developer — Orbotech Ltd., Rehovot

1996 – 1997

Computational geometry algorithm development.

EDUCATION

M.Sc. — Artificial Intelligence & Machine Learning Hebrew University, Jerusalem · 1997–2004 - Direct PhD track; research closed as Master's Thesis - Recipient of **Best Paper Award**, Uncertainty in Artificial Intelligence (UAI), 2001

B.Sc. — Mathematics and Computer Science (*Graduated with Distinction*) Ben-Gurion University, Beer-Sheva · 1994–1997

PUBLICATIONS

- **"Multivariate Information Bottleneck"** — *Uncertainty in Artificial Intelligence, 2001* 🏆 Best Paper Award (*The Information Bottleneck framework is foundational to understanding representation learning in modern deep networks*)
 - **Thesis:** "Mutual Information Extraction"
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LANGUAGES

Hebrew (native) · English (professional) · Spanish (intermediate)

INTERESTS

AI, mathematics, free software (GNU/GPL advocate, Wikipedia contributor), karate, hiking.